NEW YORK INSTITUTE OF TECHNOLOGY

Simulation of Dynamic Wireless Charging Roads in NYC

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Abstract

The number of EVs on the road in the U.S. is projected to be roughly 78.5 million by 2035, up from 4.5 million in 2023. This increase in EVs demands that a larger number of stationary wired chargers to meet the charging demands. Based on the 2035 projection, 42.2 million wired chargers would be needed. Chargers that occupy valuable urban space and add congestion around charging stations. We study the potential of dynamic wireless power transfer (DWPT) roads in NYC. To demonstrate and evaluate this system, we created a simulation in which DWPT lanes are deployed throughout NYC Midtown. The model calculates the charge an EV gains from the lanes during each trip, and it also assesses construction costs and overall efficiency metrics.

Background

- Within 2025, the rising popularity of electric vehicles (EVs) will bring upon new challenges, mainly the high electricity costs that are predominantly high rates during peak hours and the time it takes to fully charge a vehicle. Urban areas like Manhattan have many added difficulties due to limited space for public chargers, actively forcing drivers to travel out of their way just to find available EV charging stations. The SAE J2954 standard is moving system enables EVS to charge wirelessly while moving or idling, to further eliminate the need for fixed charging locations, and further reduce energy-related stress for high-traffic environments.
- Our team focused on studying the performance between between vehicle receivers (Va's) and Ground Assemblies (A's) in SAE J2954-based wireless charging systems. We study three coil configurations: Dual Receiver DD (Dual WPT3), Segmented DDQ WPT-3 and Large Single Coil WPT4. Each design was evaluated based on its energy transfer rate, efficiency, and cost.

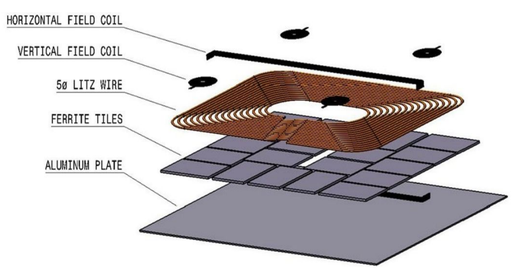

Figure 1: Ground Assembly structure defined by the SAE J2954 Standard

- Real-world deployments by companies such as Siemens, Electreon, and WiTricity are already validating this approach. For instance, Electreon has implemented dynamic wireless charging in Stockholm and Detroit, providing large-scale feasibility.
- WiTricity's market solutions are being used by automotive leaders such as BMW and Hyundai. These systems use standard protocols that are inspired by the SAE J2954 standard, to ensure efficiency.

System Outcomes

Block Diagram

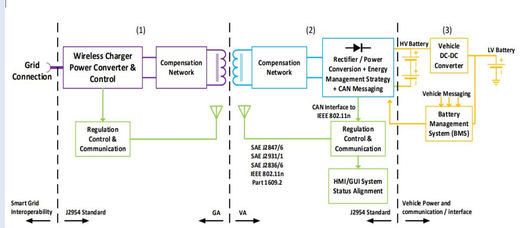

Figure 2: Block Diagram of the DWPT System Defined by the SAE J2954 Standard

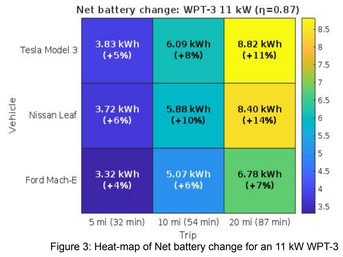

Figure 3: Heat-map of Net battery change for an 11 kW WPT-3

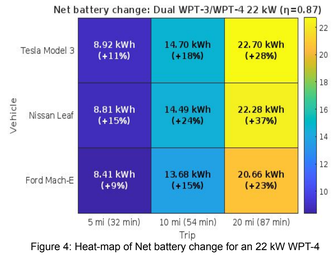

Figure 4: Heat-map of Net battery change for an 22 kW WPT-4

- Net gain 3.3 – 8.8 kWh (4 – 14 % SOC) across 5-20 mi city trips (Tesla 3, Leaf, Mach-E).
- Energy delivery dominated by stationary dwell (≈50 % while stopped), easing grid transients.
- Demonstrates that even the lowest SAE power class can offset typical urban energy use and cut public fast-charger visits by >60 %.

- Net gain 8.4 – 22.7 kWh (9 – 28 % SOC) across 5-20 mi city trips (Tesla 3, Leaf, Mach-E).
- Provides energy-neutral daily driving with only 8-20 mi of roadway, reducing battery cycling and extending pack life.
- Higher power classes are therefore recommended for bus routes, rideshare corridors, and logistics spines where throughput and dwell are predictable.

SUMO Simulation

SUMO is an open source and continuous multi-modal traffic simulation designed to handle large networks. Our goal with SUMO is to simulate the potential charging impact WPT-3 DWPT lanes can have throughout NYC Midtown (0.75 mi radius). We simulate this impact on EVs within NYC Midtown under two scenarios: WPT-3 DWPT lanes on all roads and WPT-3 DWPT lanes on optimal roads (congested roads). For our procedure, we will simulate the aforementioned EVs (Tesla Model 3, Nissan Leaf, Ford Mach-E) and calculate their resulting charge using the WPT-3 kWh per vehicle.

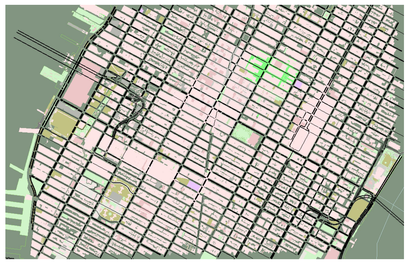

Figure 5: SUMO Simulation Map of Midtown Manhattan NYC

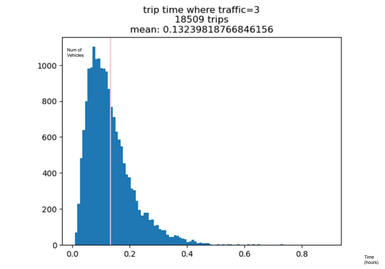

Figure 6: Histogram of time spent on road charger for high traffic volume

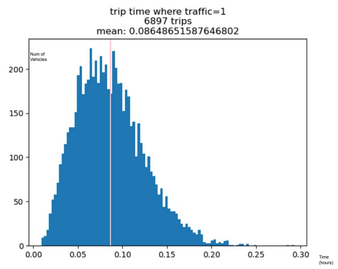

Figure 7: Histogram of time spent on road charger for low traffic volume

Cost Analysis

- We compare the cost impact of three wireless EV charging coil configurations, the WPT-3, DUAL WPT-3 and the Large single Coil WPT4, based on their size and quantity per mile. Both the DD and DDQ systems use meter coils, requiring 1609 units per lane and posting between $482,700 - $965,400 (DD) and $63,150-$1,126,300 (DDQ). In contrast, Large single Coil spans 2 meters, needing only 805 per unit per mile, which reduces the amount cost to $322,000- $644,000 per mile per lane despite the per unit pricing. This highlights the WPT4's coil clear economic advantage towards a large-scale urban deployment.
- Citywide implementation from Midtown to Downtown Manhattan (10 miles, 4 lanes would cost: $19.3 B-$38.6 B for Dual WPT3, $22.5B 45.0B for WPT3 DDQ, and $12.9 B-$25.8 B for WPT4. The WPT4 requires half the number of cells, making it the most cost-efficient. Annual power and maintenance add $500M-$1B for regulatory electricity in kWh.

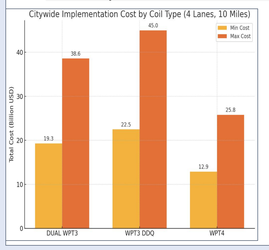

Figure 8: MATLAB Simulation showing the Citywide Implementation cost by Coil Type for 10 miles in Manhattan with $/ lane.

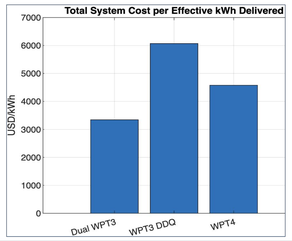

Figure 9: the total system cost per effective kWh delivered and it shows that the WPt3 DDQ would cost the most in USD for the cost rate in kWh in electricity.

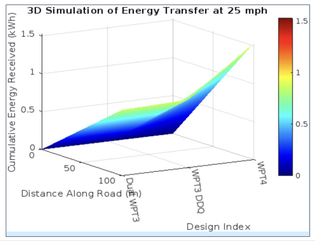

Figure 10: 3D Simulation. The left explains the energy transfer of each coil at 25 MPH from the energy received in kWh vs. distance traveled in meters.

Discussion

- It is evident that the inclusion of DWPT lanes, particularly in optimal locations, can be a great benefit to charge EV's without taking up urban real-estate or contributing to congestion.
- Further research is needed to determine if infrastructure can handle DWPT lanes electricity demands.
- SAE J2954 defines interoperable 11 kW (WPT-3) and 22 kW (WPT-4) classes for light-duty EVs, while draft J2954/3 extends the protocol to dynamic lanes and heavier vehicles, paving the way for mixed-fleet compatibility.
- Up-front installation cost, winter salting, EMAG compliance and foreign-object detection all require further R&D. Modular, prefabricated coil sections and advanced asphalt encapsulation are active research areas aimed at lowering lifetime O&M.
- By freeing sidewalk real estate and shortening dwell at stationary chargers, DWPT can improve streetscapes, yet funding strategies must ensure deployment reaches outer boroughs and disadvantaged communities, not just high-traffic business districts.

Conclusion

Continuous DWPT lanes can fully offset—and at 22 kW even exceed—the energy used on typical 5- to 20-mile midtown trips, enabling energy-neutral commuting without driver behavior change. Because charging happens in-lane, DWPT frees valuable curb space and avoids congestion linked to the tens of millions of wired chargers forecasted for 2035. Using SAE J2954 WPT-3/WPT-4 standards ensures fleet-wide compatibility, while cost modeling shows lane energy prices rival curbside DC fast charging when amortized over 15 years. Targeted deployment along high-traffic corridors positions New York City to lead the next wave of seamless, in-motion EV charging infrastructure.

Future Work

Building on our simulation results, the next phase shifts from proof-of-concept analytics to driver-facing services. Our goal is to create a mobile application that combines Google Maps routing with a live database of Dynamic Wireless Power Transfer (DWPT) lanes and EV specifications. The app will predict net State-of-Charge (SOC) change, charging cost, and carbon savings for any trip through New York City, and eventually nationwide. In parallel, we will refine the underlying infrastructure and analytics to move from research track to pilot deployment.

References

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